Lecture 0: Course Introduction

Lecturer

Tran Nguyen Lan

CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

Course Overview & Vision

PILLAR A

Thinking

Framing economic problems, understanding statistical compression, sampling distributions, and choosing fitting analytical models.

PILLAR B

Interpretation

Reading complex visualizations, statistical significance vs practical significance, and evaluating quantitative claims critically.

PILLAR C

Execution

Working directly with empirical datasets using Excel & Python to conduct regression, estimation, and executive reporting.

Pillar A: Framing the Problem

- **CLO1 — Data-to-Knowledge Chain:** Locate statistics in the *Data* → *Information* → *Knowledge* chain; explain why a statistic is a compression of information and identify what details are lost.
- **CLO2 — Sampling & Central Limit Theorem:** Distinguish a population parameter from a sample statistic; master sampling distributions and the Central Limit Theorem (CLT) as the cornerstone of valid statistical inference.
- **CLO3 — Method & Data Identification:** Given a business problem with data, correctly identify the question type (description, comparison, association, or forecasting), the data structure, and the appropriate family of methods.

Pillar B: Reading & Judging Results

- **CLO4 — Chart Selection & Visual Pitfalls:** Select suitable charts by data type and question; detect non-linearity, outliers, and hidden groups, while recognizing misleading graphics.
- **CLO5 — Statistical Significance & Inference:** Interpret confidence intervals, p-values, and regression coefficients accurately (including what they do *NOT* imply); separate statistical from practical significance.
- **CLO6 — Critical Evaluation of Quantitative Claims:** Evaluate claims for correlation vs. causation traps, omitted variables, violated assumptions, selective reporting, and over-generalisation.

Pillar C: Working with Data

CLO7 — Data Preparation:
Prepare an analysis-ready dataset and describe it thoroughly in a standard empirical summary table.

CLO8 — Applied Analytics in Excel & Python:
Execute data visualization, interval estimation, hypothesis testing, and multiple regression; effectively communicate findings to a non-statistical business audience.

Required Learning Materials

Source / Framework	Resource Specification	Core Focus Area
Primary Textbook	Wooldridge (2025) Introductory Econometrics: A Modern Approach, 8th Edition	Main reference for regression analysis and empirical modeling.
Additional	Charles Wheelan. Naked Statistics	Core intuition for real-world statistical reasoning.
Additional	Darrell Huff. How to Lie with Statistics	Classic guide on spotting misleading graphics, bias, and manipulation

Empirical Data Sources

GSO Vietnam

Tổng cục Thống kê
gso.gov.vn

Macroeconomic & demographic national data.

World Bank WDI

World Development Indicators
data.worldbank.org

Global economic indicators & social statistics.

HOSE / HNX

Financial Markets
Stock Exchanges

Capital market trading data & corporate reporting.

Kaggle & OWID

Open Datasets
Our World in Data

Global trends, social data & machine learning sets.

Course Schedule & Phases

Phase	Sessions	Role
1 — Data Foundations	1–3	From raw data to information: identify it, describe it, see it
2 — Probability Foundations	4–6	Why generalising from a sample is legitimate. Culminates in the CLT
3 — Statistical Inference	7–11	From describing a relationship to a conclusion that has been tested
4 — Modelling and Synthesis	12–14	Multiple regression and time series

Course Instructors

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Course Evaluation Criteria

